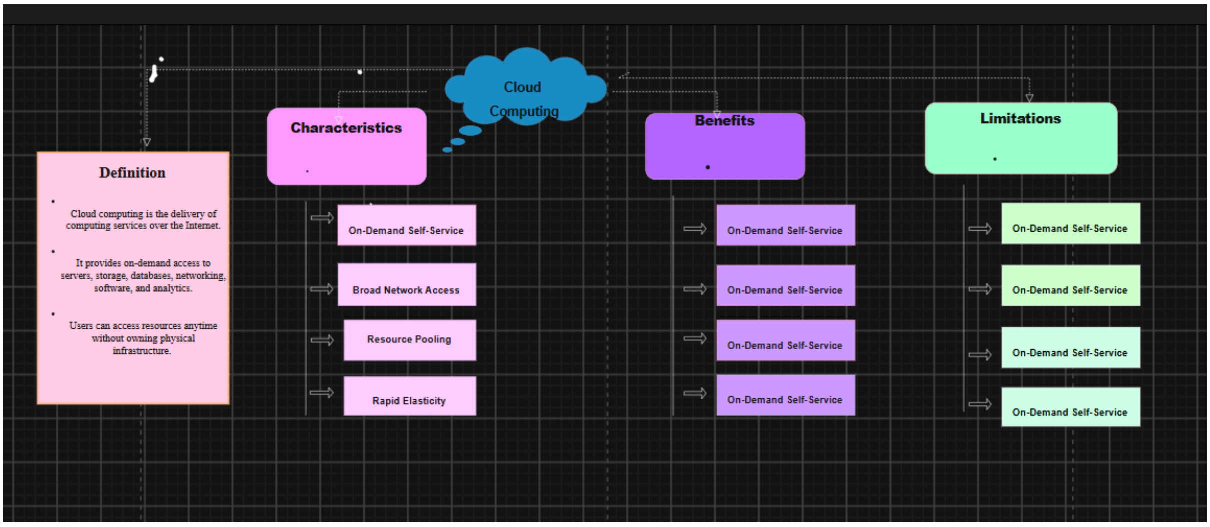
1.



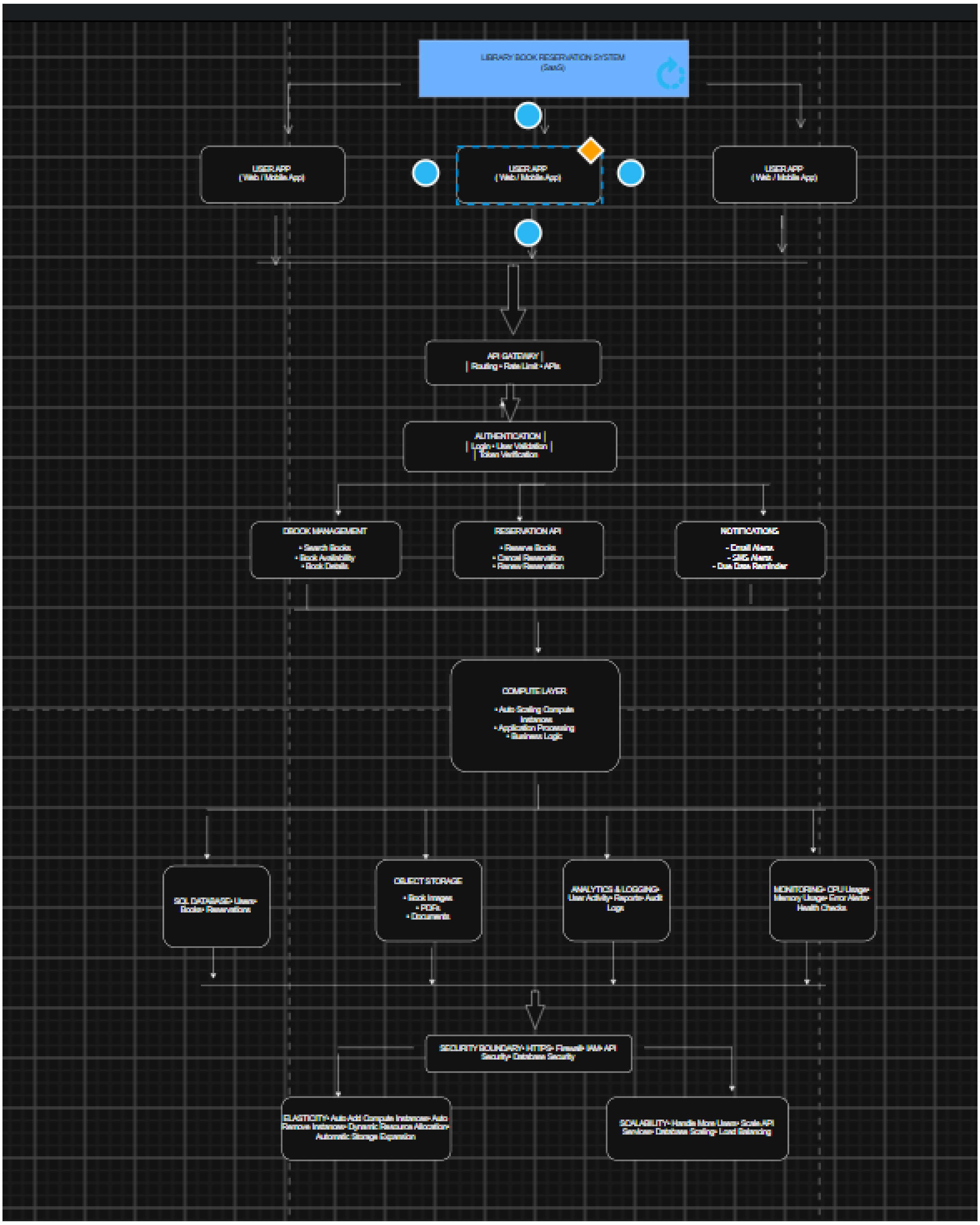
2(a)

Service Category	Cloud Provider Manages	Student Team Manages	Why This Split Fits the Project
SaaS	Software, hosting, updates, security	Users, settings, data	Focus on using the application.
PaaS	Infrastructure, OS, runtime,	Application code and data	Focus on application development.
IaaS	Servers, storage, networking	networking OS, applications, security, data	Provides greater control over resources.
FaaS	Infrastructure, OS, runtime,	]Functions and business logic	No server management required.
DBaaS	Database, backup, security	Tables, queries, application data	Easy database management.
STaaS	Cloud storage, backup, scalability	Files and documents	Secure and scalable file storage.
IDaaS	Authentication and user identity	User roles and permissions	Simplifies secure login.
Monitoring & Logging	Monitoring tools, logs, alerts	Analyze logs and fix issues	Improves application reliability.

2(b).

Project Module	Recommended Cloud Service	Example Service	Justification
Authentication	Identity as a Service (IDaaS)	Firebase Authentication / AWS Cognito	Provides secure user login, registration, password reset, and role-based access without building authentication from scratch
Static Web Hosting	Static Web Hosting	Firebase Hosting / AWS S3 + CloudFront / Azure Static Web Apps	Delivers the website quickly with high availability, scalability, and low maintenance.
Backend / API Layer	PaaS or FaaS (Serverless)	Google Cloud Run, AWS Lambda, Azure Functions	Hosts APIs and business logic while automatically handling scaling and server management.
Database	Database as a Service (DBaaS)	Firebase Firestore, MySQL on Azure, AWS RDS	Stores user details, book records, reservations, and transactions with automatic backup and scalability.
Object Storage	Storage as a Service	AWS S3, Google Cloud Storage, Azure Blob Storage	Stores book cover images, PDFs, reports, and uploaded documents securely and efficiently.
Analytics / Logging	Monitoring & Logging Service	Google Cloud Logging, AWS CloudWatch, Azure Monitor	Tracks application performance, errors, and user activity for easier maintenance and troubleshooting
Notification	Monitoring & Logging Service	Firebase Cloud Messaging (FCM), SendGrid, Amazon SNS	Sends reservation confirmations, due-date reminders, and system notifications through email or mobile alerts.
AI / Python Module	AI Platform / Serverless Python	Google Colab, Vertex AI, AWS SageMaker	Runs Python scripts for recommendation systems, prediction models, or analytics without managing servers.
Security	Cloud Security Services	AWS IAM, Azure Security Center, Google Cloud IAM	Provides encryption, identity management, access control, and protection against unauthorized access.
Backup	Backup as a Service	AWS Backup, Azure Backup, Google Backup	Automatically creates backups of databases and storage to prevent data loss and enable recovery.
Monitoring	Cloud Monitoring Service	Google Cloud Monitoring, AWS CloudWatch, Azure Monitor	Continuously monitors system health, server performance, uptime, and resource utilization to ensure reliable operation.

3.


LIBRARY BOOK RESERVATION SYSTEM
(SaaS)
USER APP
(Web / Mobile App)
USER APP
(Web / Mobile App)
USER APP
(Web / Mobile App)
API GATEWAY |
| Routing • Rate Limit • APIs
AUTHENTICATION |
| Login • User Validation |
| Token Verification
EBOOK MANAGEMENT
• Search Books
• Book Availability
• Book Details
RESERVATION API
• Reserve Books
• Cancel Reservation
• Renew Reservation
NOTIFICATIONS
- Email Alerts
- SMS Alerts
- Due Date Reminder
COMPUTE LAYER
• Auto Scaling Compute Instances
• Application Processing
• Business Logic
SQL DATABASE• Users• Books• Reservations
OBJECT STORAGE
• Book Images
• PDFs
• Documents
ANALYTICS & LOGGING• User Activity• Reports• Audit Logs
MONITORING• CPU Usage• Memory Usage• Error Alerts• Health Checks
SECURITY BOUNDARY• HTTPS• Firewall• IAM• API Security• Database Security
ELASTICITY• Auto Add Compute Instances• Auto Remove Instances• Dynamic Resource Allocation• Automatic Storage Expansion
SCALABILITY• Handle More Users• Scale API Services• Database Scaling• Load Balancing

4. **Technical Justification**

Service Model Selection

Software as a Service (SaaS) is selected for the Library Book Reservation System because users access the application through a web or mobile browser without installing any software. The cloud provider manages the infrastructure, platform, and application hosting, while users simply use the service online.

API Role

The **API Gateway** acts as the communication bridge between the user application and backend services. It receives user requests, performs request routing, authenticates users, manages APIs, applies rate limiting, and securely transfers data between the application, database, and cloud storage.

Elasticity

The system demonstrates **elasticity** by automatically increasing or decreasing cloud resources according to user demand. During peak library usage, additional compute instances are created automatically, and when demand decreases, unused resources are released to reduce operational costs.

Scalability

The architecture supports **scalability** by allowing the application to serve more users without performance degradation. Multiple application instances, load balancing, scalable databases, and cloud storage ensure that increasing numbers of users and reservations can be handled efficiently.

Virtualization / Managed Compute Choice

The application uses **managed cloud compute services** (virtual machines, containers, or application services) instead of physical servers. The cloud provider manages hardware resources, operating systems, virtualization, and maintenance, enabling the development team to focus only on application development and business logic.

Cloud Responsibility Boundaries

The **Shared Responsibility Model** defines the responsibilities of both the cloud provider and the application team.

Cloud Provider Responsibilities

- Managing physical servers
- Virtualization
- Networking
- Storage infrastructure
- Operating system updates
- Cloud platform availability

Application Team Responsibilities

- Application development
- User authentication
- API implementation
- Library database management
- User data protection
- Access control and application configuration